Aneesh Kadiyala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.53
Narayana Jr. College, Mehdipatnam	Class XII (Telangana State Board of Intermediate Education)	2023	94.50%
Meridian School	Class X (Central Board of Secondary Education)	2021	95.20%

EXPERIENCE

Intern in 360 Technology

- Worked on implementation and testing of Kavalan AI, an **enterprise AI gateway** that monitors network traffic and blocks requests going to unauthorized providers and offers a configurable DLP policy along with a comprehensive audit log
- Implemented a CRUD API using **Axum** and collaborated with front-end developers for the dashboard
- Implemented an **MiTMproxy addon** to perform network filtering and TLS inspection on payloads

Project Intern in WiSig Networks

- Integrated post-quantum **ML-KEM** key encapsulation to replace classical ECIES for **SUPI** concealment using **liboqs**
- Analyzed **Robust Header Compression** implementation and interface specifications for cellular stack integration

PROJECTS

Open-Source Contributions to rust-lang/rust-clippy

- **Added new lint**, empty_enum_variants_with_brackets, and fixed it to suppress diagnostics for reachable enums
- **Generalized a false-positive fix** for if_same_then_else when branches contain different comments
- **Corrected inconsistencies in the Clippy Book** regarding early and late lint passes
- **Fixed a false positive** in manual_unwrap_or_default for expressions that do not implement Default
- **Reviewed and approved a contribution** subsequently merged into Rust Clippy

Modelling Inventory Optimization as a Convex Optimization Problem | *Course: Convex Optimization*

- **Performed literature review** of existing approaches used to model inventory optimization as a convex problem.
- **Implemented and compared** the performance of algorithms proposed by the reviewed literature

DSL for Signal-Processing | *Course Project: Compilers*

- **Designed a DSL to work with vectors** and perform common operations such as dot product and 1D convolution
- Implemented it using C++, Flex, Bison and LLVM IR

Single-Image Super-Resolution Using CNN | *Course Project: Machine Learning*

- Implemented SRCNN to correct artifacts introduced by bicubic upsampling using MSE loss and the DIV2K dataset

Adaptive Noise Cancellation | *Course Project: Digital Signal Processing*

- Designed and implemented a real-time noise cancellation algorithm with full and partial suppression modes
- Demonstrated a 24 dB SNR for **2nd-order RLS** as opposed to approximately 14 dB for LMS and NLMS

CERTIFICATIONS

- Pursuing a **Minor in Machine Learning** from IIIT Hyderabad (September 2025 - present)

- NVIDIA's Fundamentals of Accelerated Computing with CUDA C/C++
- NVIDIA's Fundamentals of Accelerated Computing with CUDA Python

SKILLS

Programming Languages: Rust, Python, C

Tools: MATLAB, Verilog, LaTeX

Libraries: NumPy, PyTorch, JAX, Tokio

LLMs & Agentic Systems: Multi-Agent Workflows, Context Window Optimization, Tool Calling

Miscellaneous: Design Patterns

RELEVANT COURSES

Computer Science: Compilers, DBMS, GPU Programming, Distributed AI Training on Large-Scale GPU Clusters

AI: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Reinforcement Learning

Math: Numerical Methods, Linear Optimization, Convex Optimization, Statistical Inference

Communications and Signal Processing: Digital Signal Processing, Communication Systems, Information Theory

Quantum Computing: Introduction to Quantum Computing, Quantum Error Correction

POSITIONS OF RESPONSIBILITY

Internship Coordinator in Office of Career Services

- Acted as a point of contact between companies and students
- Helped conduct various Office of Career Services events like Japan Day

Software Engineering Core in Tinkerers' Lab

- Conducted hands-on introductory sessions on Linux and Git for juniors
- Assisted in the development of webpages for various events conducted by Tinkerers' Lab
- Assisted in smooth progression of various Tinkerers' Lab events

EXTRACURRICULARS

- NSS (National Service Scheme) Volunteer